Beyond the edge of stability: periodic windows, quasi-periodicity and phase locking in the learning-rate plane of gradient descent

Authorship to be finalized by the human researcher

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Abstract

Full-batch gradient descent with a large learning rate can oscillate, cycle or behave chaotically instead of settling. Trainability maps over learning-rate planes record a binary convergent-or-divergent label, which conflates these dynamical regimes. We resolve the dynamics behind such maps and organize the results around one thesis: beyond the edge of stability, the period-2 oscillation gives way to quasi-periodic training and phase-locked periodic windows.

(1) Confirmed in a prospectively designed, independently audited study of two small tanh networks with separate learning rates for their two layers: the learning-rate planes contain 61 numerically certified attracting periodic orbits (periods 4 to 58) embedded in rerun-confirmed chaos, and 87.1% to 99.97% of the cells a published binary convergence score labels convergent were not classified as fixed points after 20,000 updates.

(2) Exploratory: the edge-of-stability period-2 orbit loses stability along a traced curve where a complex pair of two-step multipliers crosses the unit circle. The first Lyapunov coefficient is negative (supercritical) at 62 of 69 points, and in 8 of 8 dynamical checks the behaviour matches its sign. Just beyond the curve the measured rotation number equals the crossing angle to a median 5.8×10^{-4} and 3.6×10^{-4} in two maps. The same crossing appears in a 529-parameter network trained on handwritten digits with one learning rate (predicted rotation 0.0077528, measured 0.007751), and in the two-mode linear network whose rotational edge was derived by Hofmann, where it is supercritical with locked windows beyond it.

(3) Not established: under frozen rules, the ancestry test of a rule-selected window returned "alternative ancestry" (post hoc, its locked region narrows with exponents 1.518 and 0.513, against 1.5 and 0.5 for a resonance tongue); locking was not resolved in the real-data network; and locking and rotation features added no robust predictive value beyond the Lyapunov exponent and sharpness.

Takeaway: the local Neimark–Sacker prediction is supported in these models; that the periodic windows descend from its resonances is not yet established. Nothing here concerns stochastic or large-scale training or establishes a training benefit.

Study status. Results labeled CONFIRMED (PROSPECTIVELY DESIGNED, INDEPENDENTLY AUDITED) come from the map study of Section 4, whose design, thresholds and analysis were pushed before outcomes and repeated by an independent audit. Results labeled EXPLORATORY (Section 5) were found while exploring; some used acceptance rules frozen before the run, and we say so where they did. Provenance for every figure and number is described in Section 9.

1 Introduction

Training with a learning rate beyond the classical stability limit is common practice. At the edge of stability, the sharpness of the loss hovers near $2/\eta$ while the loss still decreases over long horizons [6]; self-stabilization explains how sharpness is held there [9], progressive sharpening how it gets there [1], and the catapult phase what happens to the loss at somewhat larger rates [21]. Simplified models show period doubling and chaos as the step size grows [5, 13, 18]. Over two-dimensional hyperparameter planes, the boundary between trainable and divergent settings is fractal [22, 23, 28, 31]. These maps record whether the loss falls and stays bounded, not what the parameters do.

We ask what the dynamics inside such maps are. We treat full-batch gradient descent with separate learning rates for the hidden and output layers as a two-parameter family of maps on parameter space. For every cell of a learning-rate grid we screen the parameter state at the end of a long run, certify selected periodic orbits with Newton's method on the exact Jacobian, and follow orbits and their stability boundaries through the plane.

Thesis and status. We organize the results around one thesis: beyond the edge of stability, the period-2 oscillation gives way to quasi-periodic training on an invariant circle, rational rotation numbers lock into periodic windows, and the windows of the learning-rate plane are largely these locked orbits. Each result below says exactly what passed. The certified windows and the comparison with the binary score are confirmed. Exploratory analyses support the local part of the thesis: the period-2 orbit acquires complex multipliers along a curve, the crossing is supercritical on most of it with dynamics that match the normal form, quasi-periodic training starts with a rotation equal to the crossing angle, and the same crossing appears in a real-data network and in a linear network. The global part is not established: a frozen test did not confirm that a rule-selected window descends from its resonance, and no prespecified path into sustained chaos has been tested. We call the crossings Neimark–Sacker candidates except where the nondegeneracy coefficient and dynamical checks were computed.

Prior work. Periodic windows, period doubling, positive Lyapunov exponents and broadband power spectra in learning dynamics, with the learning rate as bifurcation parameter, date back at least to van der Maas et al. [32], who studied a three-unit autoassociator. Their delay-embedded chaotic attractors are thin folded curves, effectively one-dimensional chaos of the period-doubling kind, unlike the invariant circles of the two-step map studied here. Period-colored maps over optimizer hyperparameters have been drawn from a two-dimensional projection of the error, without full-state verification [30]. Hofmann [13] studies finite-step gradient descent as a dynamical system in solvable models: the edge of stability as the first (flip) bifurcation of the training map, the period-doubling route with windows, and, in a two-mode depth-two linear network, a "rotational edge" beyond which the unstable multipliers of the aligned 2-cycle are complex and the system saturates in a persistently misaligned oscillatory state. The edge is Hofmann's result; we show, in that model and in nonlinear networks, what lies beyond it: a supercritical Neimark–Sacker bifurcation with an invariant circle whose rotation is set by the crossing angle, and locking. Riddled basins and learning-rate-dependent reproducibility are documented by Ly and Gong [24]. Invariant curves and phase-locking Arnold tongues appear in a multiplicative-weights learning rule for population games [25], and chaos in learning a two-person game goes back to Sato et al. [27]; both concern games rather than descent on one loss. Heavy-ball momentum has cycles that rotate by $1/K$ [12].

This paper's contribution.

- Confirmed: 61 certified attracting windows embedded in chaos in the learning-rate planes of two small networks, repeated by an independent audit, and a cell-by-cell comparison showing that the published binary score conflates fixed points, periodic orbits, chaos and drift.
- Exploratory, local mechanism: a traced curve of complex-pair crossings of the period-2 orbit; its first Lyapunov coefficient, generalized Neimark–Sacker points and dynamical checks; rotation-number maps in which quasi-periodic training begins at the crossing angle.
- Exploratory, generality: the same crossing with a quantitatively predicted rotation in a real-data network with one learning rate; a supercritical Neimark–Sacker bifurcation with locking beyond the rotational edge of a linear network; coexisting period-2 branches with their own crossings in a 72-parameter network.
- Reported negatives: a frozen ancestry test that did not pass; no locking resolved in the real-data network; no robust incremental value of rotation information for reproducibility or loss.
- Code, receipts and a verification script that re-derive every figure, table and number.

Not claimed.

- No route to chaos: no prespecified connected path through sustained chaos has been tested.
- No proof that the windows are Arnold tongues: the Birkhoff test is a diagnostic of projected orbits, and the frozen ancestry test did not pass.
- No claim about stochastic, minibatch or large-scale training, or other architectures and data beyond those shown, and no training benefit.

2 Gradient descent as a family of maps

Let $L(\theta)$ be the full-batch training loss of a network with parameters $\theta \in \mathbb{R}^n$, split into hidden-layer and output-layer blocks. Gradient descent with per-layer learning rates is the map

$$\theta_{t+1} = F(\theta_t) = \theta_t - P\,\nabla L(\theta_t), \qquad P = \mathrm{diag}(\eta_h \mathbf{1}_{\mathrm{hidden}}, \eta_o \mathbf{1}_{\mathrm{output}}), \tag{1}$$

a family of maps indexed by the two rates (η_h, η_o). Its Jacobian is $DF(\theta) = I - PH(\theta)$ with H the Hessian of L. An orbit of period p is stable when all eigenvalues (multipliers) of $DF(\theta_{p-1}) \cdots DF(\theta_0)$ lie inside the unit circle. Generically it loses stability when a multiplier crosses the circle: at $+1$ (fold), at -1 (flip, which doubles the period) or as a complex pair $e^{\pm 2\pi i\omega}$ (Neimark–Sacker). Under transversality and a nonzero first Lyapunov coefficient d, a Neimark–Sacker bifurcation creates an invariant closed curve: attracting with amplitude growing like the square root of the distance beyond the crossing when $d < 0$ (supercritical), repelling and present before the crossing when $d > 0$ (subcritical) [20].

Why the candidates start at period 2. $DF(\theta)$ is similar to the symmetric matrix $I - P^{1/2} H P^{1/2}$, so at a fixed point all multipliers are real and a fixed point loses stability by fold or flip only. The edge-of-stability oscillation is the period-2 orbit born at that flip. For a period-2 orbit $\{\theta_a, \theta_b\}$, the two-step Jacobian is similar to $S_b S_a$ with $S_j = I - P^{1/2} H(\theta_j) P^{1/2}$. A product of two symmetric matrices has real eigenvalues when one factor is positive definite, but at the edge of stability each

S_j has a negative eigenvalue in the oscillating direction, so complex multipliers become possible. This is an elementary observation; complex unstable multipliers of an edge-of-stability 2-cycle were derived for a two-mode linear network by Hofmann [13].

Circle maps and phase locking. On an invariant circle the dynamics has a rotation number ρ, the mean fraction of a turn per iterate. When $\rho = m/q$ is rational the motion can lock onto a periodic orbit; the locked parameter sets are Arnold tongues, rooted at the points of the Neimark–Sacker curve where the crossing angle is m/q, with widths that grow like the distance beyond the curve to the power $(q-2)/2$; rotation numbers along a parameter path form a devil's staircase, and the widest plateau between two locked rationals is typically their Farey mediant [2, 14, 15]. Breakdown of the circle leads to chaos [3, 8, 26]. Because the circle in question belongs to the two-step map, a locked orbit at rotation m/q has period $2q$ for F; this relation is a consequence of the setting, not evidence.

Shrimps. In two-parameter families of dissipative maps, periodic windows inside chaos often take a characteristic "shrimp" shape bounded by fold and flip curves [10, 11]. We test for this anatomy explicitly rather than assume it.

3 Methods

Network	Parameters	Training points	Architecture	Weight decay	Seeds	Maps
S1	72	72 standard-normal	8 inputs, 8 tanh units, no biases, mean-field	0	28001, 28002	2 coarse
S1R	72	72 standard-normal	8 inputs, 8 tanh units, no biases, mean-field	10^{-3}	28001, 28002	2 coarse, 2 zoom
S2	13	24 from a tanh teacher	2 inputs, 3 tanh units, biases	10^{-3}	28011, 28012	2 coarse, 2 zoom

Table 1: Design of the map study. Every map has 256×256 logarithmically spaced pairs of hidden- and output-layer rates; coarse maps span 3 decades in each rate and zoom maps 0.36 decades. Every cell runs 20,000 full-batch updates in float64 from its seed's initial parameters. S1 is the published configuration of the fractal-trainability study and is reported descriptively.

Networks and data. Table 1 lists the design of the map study. S1 reproduces the published mean-field configuration of the fractal-trainability study [28]; S1R adds weight decay; S2 is a 13-parameter tanh regressor fitted to points from a random tanh teacher. We refer to a network and seed together, for example S2-28011. Training is deterministic full-batch gradient descent in float64.

Three different quantities. The published binary score measures loss reduction: a run counts as convergent when its loss, normalized by the initial loss and averaged over the last 20 of the first 1,000 updates, is below one [28]. Our screening labels classify the parameter state at the end of a finite run. Certification validates individual periodic orbits. The three need not agree, and we compare them cell by cell.

Screening. Every cell runs 20,000 updates from the seed's initial parameters. The last 384 updates are screened for the smallest full-state recurrence period up to 64: a period p requires the state to recur with componentwise tolerance $10^{-8} + 10^{-6}\,|\theta|$ over the whole tail, which spans at least 6 repeats. The largest finite-time Lyapunov exponent (FTLE) is averaged over the last 4,000 updates after 1,000 updates of tangent alignment. Labels are fixed point, periodic, chaotic (FTLE above 0.005 per update), drift (parameter norm still growing by more than 1% over the FTLE window), unresolved, or divergent (non-finite or larger than 10^8).

Numerically certified. An orbit of period p is numerically certified when damped Newton iteration on $F^p(\theta) - \theta = 0$, using the exact chain-rule Jacobian of F^p, reaches an undamped relative correction below 10^{-10} within 60 iterations; the converged point closes to within $10^{-9} + 10^{-8}\,|\theta|$ in every component; it lies within 10^{-3} (relative) of the screened candidate; every proper-divisor iterate is separated from it by at least 10^{-6} (relative), so the period is primitive; and it is attracting when the spectral radius of the monodromy matrix is below $1 - 10^{-7}$. This is a floating-point verification, not a computer-assisted proof.

Embedded windows and audit. A connected component of equal-period cells qualifies as an embedded window when its period is at least 3, it has at least 4 cells, no cell of its exterior ring is a fixed point or has a proper-divisor period, at least 50% of the ring is rerun-confirmed chaos (reruns of 30,000 updates with a 10,000-update FTLE and a different tangent vector, up to 48 ring cells per component), and its representative orbit is certified and attracting. The audit was a separate program that imports neither the batched engine, the orbit certifier nor the selection helpers: it rewrote the one-cell map, the recurrence screen, the zoom rule and the component and ring rules, rechecked every certificate with finite-difference Jacobians (all 154 of 154 pass closure, primitivity, stability and seed checks; spectral radii agree within 1.7×10^{-7}), re-simulated the ring cells of every qualifying component, recomputed the gate on its own evidence, and found 0 mismatches among the 276 receipt files.

Continuation, boundary type and criticality. From the center of a window, its certified orbit is continued along 64 rays in the rate plane until it loses stability, located by bisection, and the boundary is typed by the crossing multiplier: fold ($+1$), flip (-1) or complex pair. The same procedure traces where the period-2 orbit loses stability along slices of fixed output rate; several traces are merged into one curve. At each point of that curve we compute the first Lyapunov coefficient d of the Neimark–Sacker normal form with Kuznetsov's formula for n-dimensional maps, using the exact dense two-step Jacobian and Richardson-extrapolated differences of exact Jacobian-vector products for the higher derivatives; the implementation reproduces the explicit planar formula on random planar maps.

Rotation numbers and the Birkhoff test. For a non-periodic cell we take the two-step samples $z_k = \theta_{2k}$, project them on their first two principal axes, and define the rotation number as the mean unwrapped angle increment in turns per two steps. Rotation-number maps run this on every cell of a grid (float64 on a GPU, 20,000 screening updates followed by 6,000 two-step samples). For a certified orbit of period p we take its two-step points ($q = p/2$ points for even p, $q = p$ for odd p), sort them by angle in their principal plane, and say the orbit passes when every step advances the angular rank by the same shift m with $\gcd(m, q) = 1$, the ordering of a Birkhoff orbit of rotation m/q. The number of points q, and therefore the relation $p = 2q$, is fixed by the period; what the test measures is the ordering and the shift. Orbits with 2 or 3 points pass by construction. The

null is the pass probability of the same test for uniformly random angles; for a group of orbits we report the expected number of chance passes. The refined test also orders points along the shortest closed tour through them and tests the points of $F^{2^{k+1}}$ ($k = 1, 2, 3$), which can detect orbits born by period doubling inside a locked window; it counts only tests on at least 5 points, and its null for each orbit is the sum of the per-test pass probabilities (a union bound), so the summed null is an upper bound on the expected number of chance passes.

Resonance ancestry. For a window selected by a fixed rule (the 1/5 window of S2-28011), we locate the points of the crossing curve whose angle is 1/5, check the onset there, and follow the locked period-10 interval on six cross-sections at distances $0.0128/4^k$ beyond the curve, finding both of its fold edges on each. Acceptance rules were frozen before the network run: controls must pass first; the interval must be fold-bounded on every level with shrinking width and amplitude; and a linear fit of its center over all levels must extrapolate to within 10^{-4} decades of the predicted point. A second stage tries to connect the tracked interval to the window.

4 Periodic windows inside chaotic training

Status: CONFIRMED (PROSPECTIVELY DESIGNED, INDEPENDENTLY AUDITED).

The learning-rate planes of S1R and S2 contain 61 embedded windows that qualify on both the producer's and the auditor's evidence (62 and 63 respectively; the disagreements all sit at the 50% ring threshold). All 154 attempted orbit certifications verified. Confirmed periods range from 4 to 58 and include the odd periods 5, 11, 13, 17 and 29. The exterior rings were checked with 2,801 reruns, and 157 of 160 randomly chosen chaotic cells (16 per map) stayed chaotic on a rerun with a different tangent vector.

Figure 1 compares these dynamics with the published binary score. In the six coarse maps, 87.1% to 99.97% of the cells the score labels convergent were not classified as fixed points after 20,000 updates; the score's success means the loss fell, not that the parameters settled. Task success and non-convergent learning dynamics can coexist: van der Maas et al. [32] reported that their network performed its task across the whole learning-rate range they studied, including where learning was chaotic. The published configuration without weight decay (S1) has 0 periodic cells in both seeds: its runs diverge, stay chaotic, or drift toward large weights while the loss slowly falls. Adding weight decay makes the windows appear.

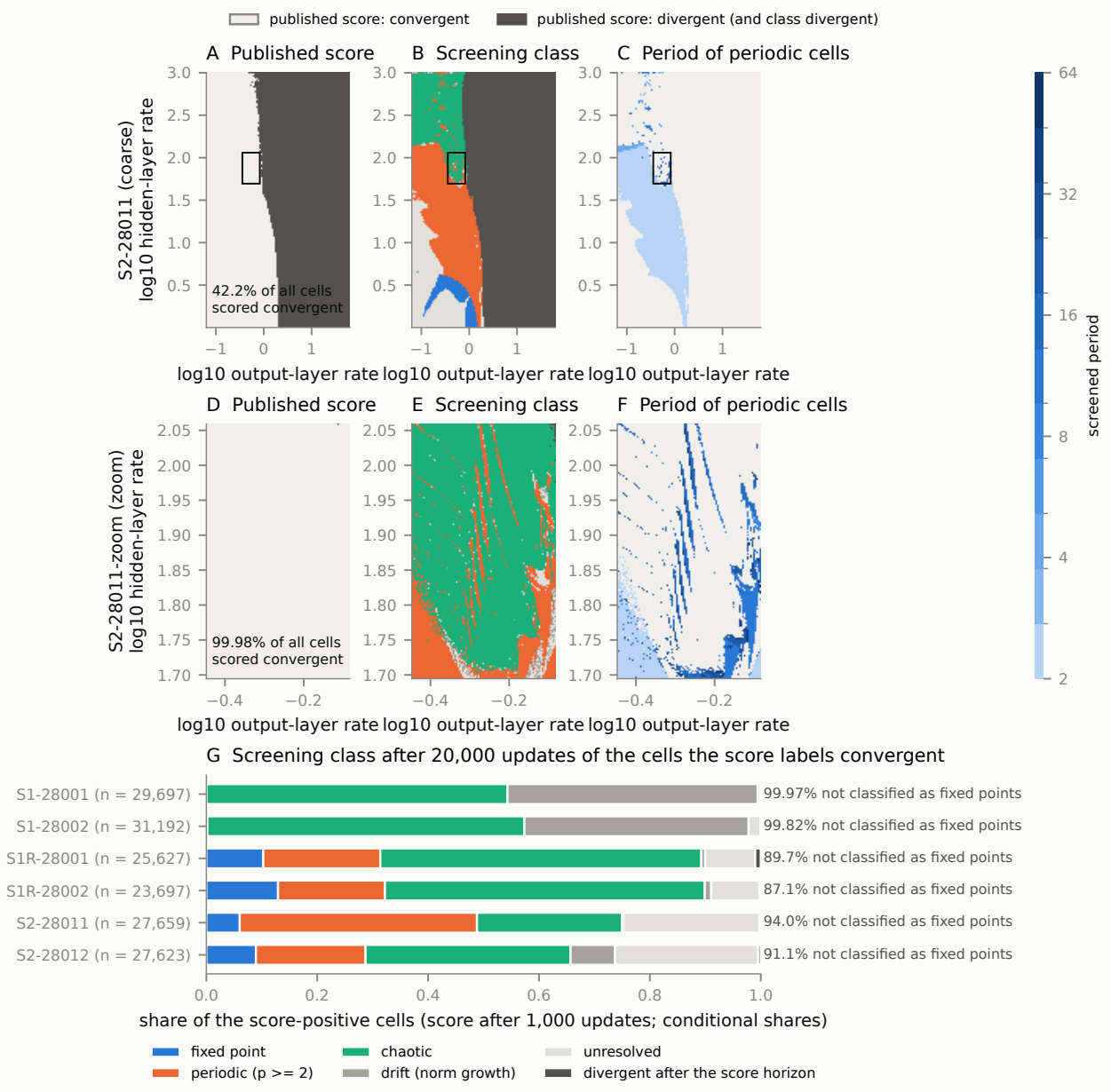


Figure 1: The binary score conflates dynamical regimes: the cells it labels convergent carry fixed-point, periodic, chaotic and drift labels after 20,000 updates. Panels A and D show the published score after 1,000 updates; the printed share is the fraction of all cells in that map scored convergent. Panels B, C, E and F show the screening class and, for periodic cells, the period of the same cells after 20,000 updates (S2-28011, coarse and zoomed; the rectangle marks the zoom). Panel G is conditional: among the cells the score labels convergent, the share in each screening class, for all six coarse maps. Screening labels are finite-time labels; only certified orbits are verified. *Generated by* `paper/gd-route/figures/f1_binary_vs_period.py`; *receipt* `figures/f1_binary_vs_period.provenance.json`.

5 Exploratory evidence for the thesis

Status: EXPLORATORY. The analyses in this section were chosen while exploring and used data that shaped the thesis, except where frozen acceptance rules are stated.

5.1 Quasi-periodic training begins where the period-2 orbit acquires complex multipliers

We continued the period-2 orbit of S2-28011 toward larger hidden-layer rates and merged several traces into one curve of 69 points along which the orbit loses stability as a complex pair of two-step multipliers crosses the unit circle; the crossing angle runs from 0.1670 to 0.2987 turns per two steps. Two rotation-number maps cover the region beyond it (Figure 2). At 0.002 decades beyond the curve, the measured rotation number equals the crossing angle at the same output rate with median absolute difference 5.8×10^{-4} over 39 points in the main map (largest 4.7×10^{-3}) and 3.6×10^{-4} over 31 points in the right-wing map (largest 2.7×10^{-3}). Of the resonance points inside each map, 5 of 7 and 5 of 6 have locked cells of the same rotation nearby; the others lie where the maps show no locked region of that rotation. In an independent staircase computation on the slice through the period-18 window, the curve places the crossing at $\log_{10} \eta_h = 1.738$ with angle 0.2180, and the first quasi-periodic cell lies at 1.738 with rotation 0.2180.

5.2 The crossing is mostly supercritical, and its dynamics match the normal form

At 62 of the 69 curve points the first Lyapunov coefficient is negative, so the crossing is supercritical: quasi-periodic training is born continuously (Figure 3). At 7 points, at both ends of the curve, it is positive. The sign changes three times, near $\log_{10} \eta_o = -0.417$, $+0.005$ and $+0.053$; these are generalized Neimark–Sacker (Chenciner) points. In 8 of 8 dynamical checks the behaviour matches the sign: at supercritical points the circle amplitude follows the square-root prediction (median relative error 0.4%, largest 5.5% next to a sign change), and at subcritical points the state jumps away beyond the curve while escape thresholds before it grow like the square root of the offset, with hysteresis. In the 72-parameter S1R network, 15 of 16 traced points are supercritical.

The fixed-start maps show quasi-periodic training beyond the supercritical arc. Where they show divergence just beyond the curve, the cause is not the bifurcation: at 7 curve points the fixed-start map is divergent even just below the curve, where the period-2 orbit is stable, and 4 of those points are supercritical, with a small attracting circle reached when training starts near the 2-cycle. The divergent arc of the fixed-start maps is a basin effect of the shared initialization, not the bifurcation.

5.3 States along one slice

Figure 4 shows what training looks like along that slice. Beyond the crossing, the two-step samples lie on a closed curve (rotation 0.2159, FTLE 0.0001); inside the period-18 window they are nine points (rotation 0.2222, FTLE -0.0072); between the 2/9 and 1/4 plateaus the curve thickens and breaks (FTLE 0.0504).

5.4 Rotation numbers form staircases

Along the slice through the period-18 window of S2-28011 (Figure 5), the rotation number falls from 0.2180 at the onset to 0.2158 at $\log_{10} \eta_h = 1.780$, then rises and locks at 2/9 for $\log_{10} \eta_h$ from 1.833 to 1.879. Of the 168 cells between the 2/9 and 1/4 plateaus, 167 are chaotic candidates; training

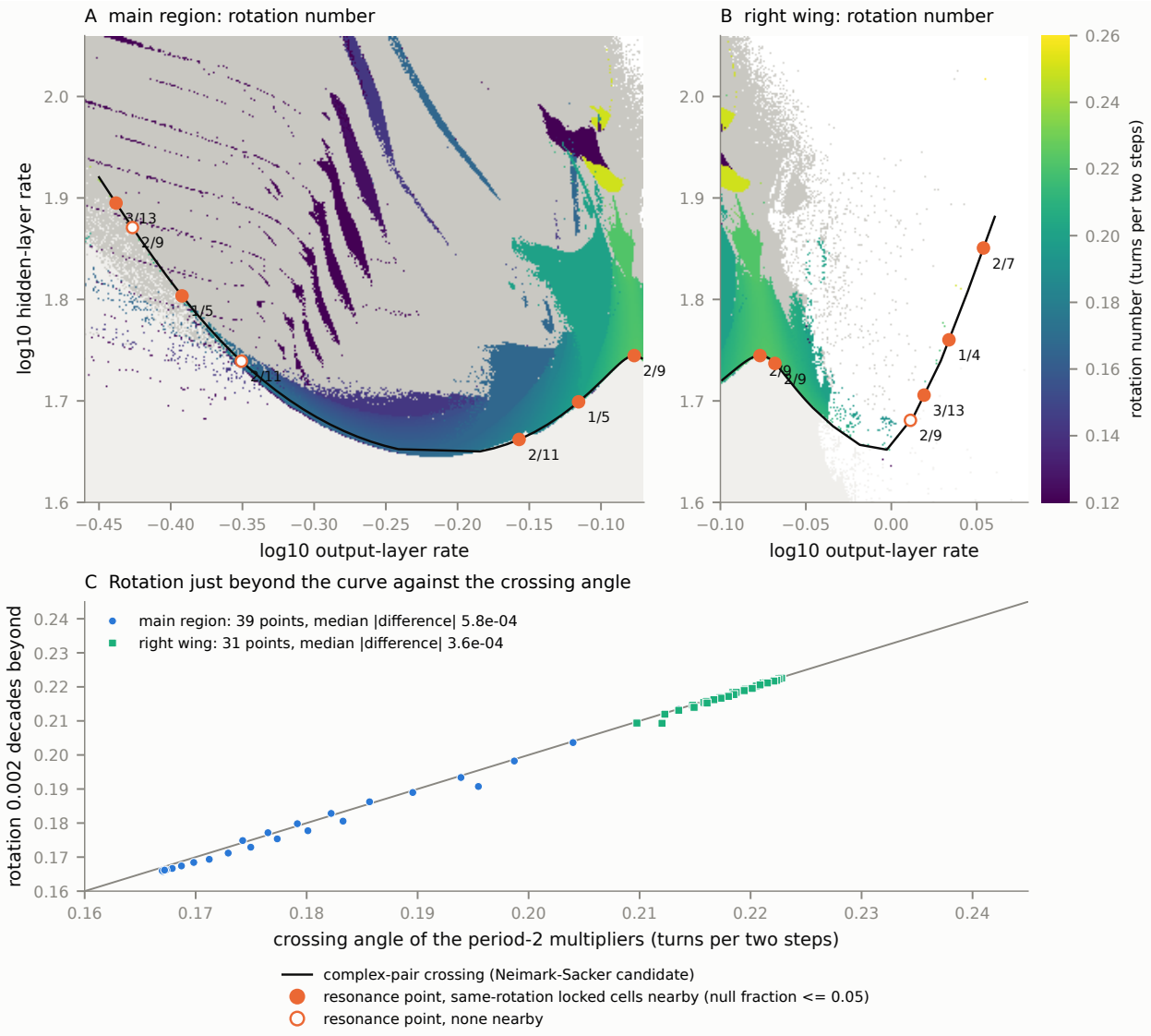


Figure 2: Beyond the curve where the period-2 orbit acquires complex multipliers, the rotation number of training starts at the crossing angle and varies continuously, and periodic windows appear as plateaus. A and B: rotation-number maps of S2-28011 (1,024 × 1,024 and 1,024 × 400 cells) with the merged crossing curve (69 points, black) and the resonance points where the crossing angle is m/q (filled when locked cells of the same rotation lie closer to the point than in 95% of random placements, hollow otherwise; the 95% level is a descriptive cut); chaotic cells gray, divergent white, period 2 light. C: rotation 0.002 decades beyond the curve against the crossing angle at the same output rate. Resonance points are interpolated between traced points. *Generated by* `paper/gd-route/figures/f2_rotation_map.py`; *receipt* `figures/f2_rotation_map.provenance.json`.

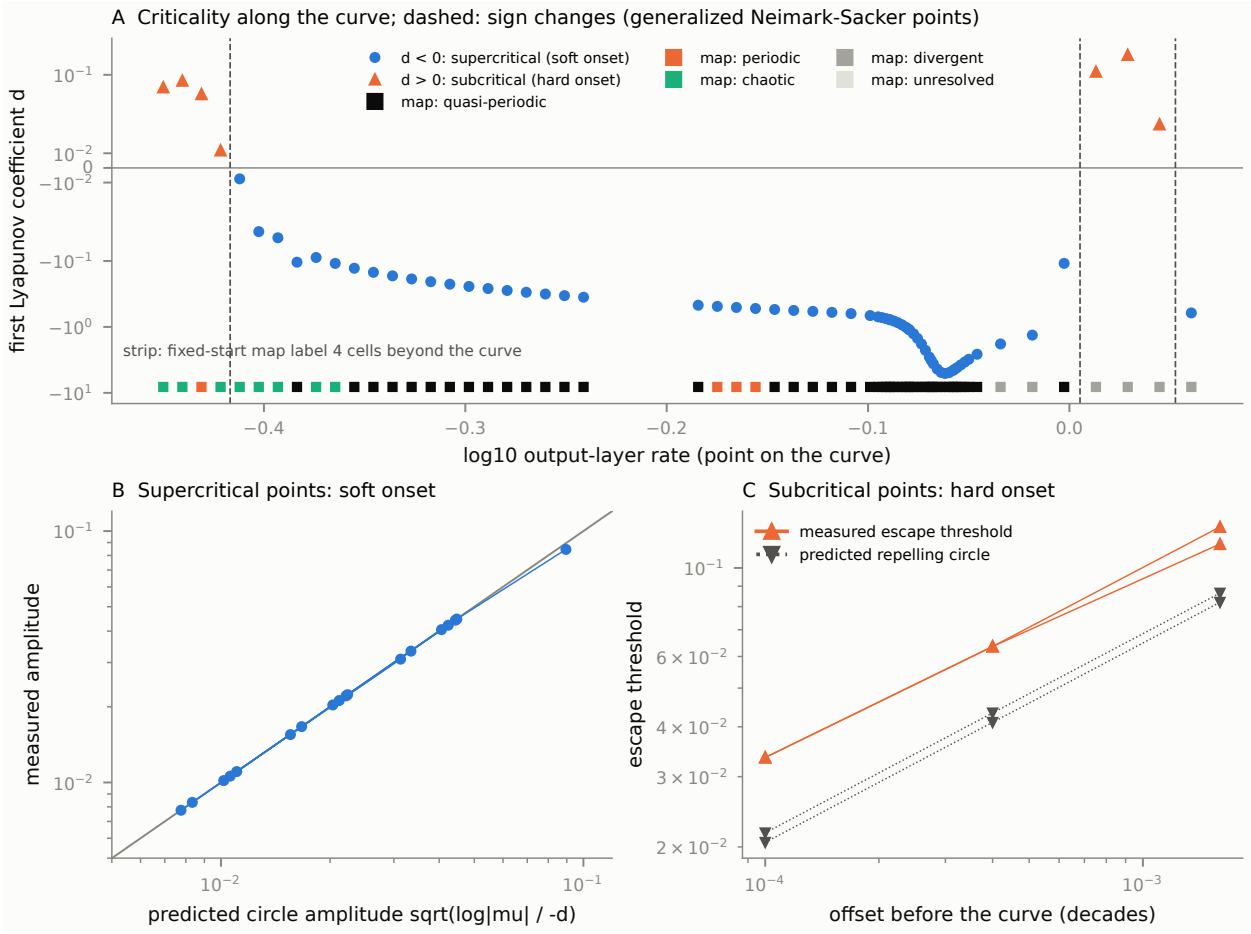


Figure 3: The first Lyapunov coefficient is negative on most of the curve, changes sign three times, and predicts how the period-2 orbit loses stability. A: d at every point of the merged curve (symmetric-log axis); dashed lines mark sign changes, located to the trace spacing; the strip shows the label of the fixed-start rotation maps four cells beyond the curve. B: at supercritical points, the amplitude of the circle reached from near the 2-cycle against the normal-form prediction, at three offsets beyond the curve. C: at subcritical points, the escape threshold before the curve against the offset, with the predicted repelling circle. A point counts as consistent when its amplitudes match within 6% (supercritical), or when the state jumps to an offset-independent distant amplitude and the escape threshold grows like the square root of the offset (subcritical). *Generated by* `paper/gd-route/figures/f7_criticality.py`; *receipt* `figures/f7_criticality.provenance.json`.

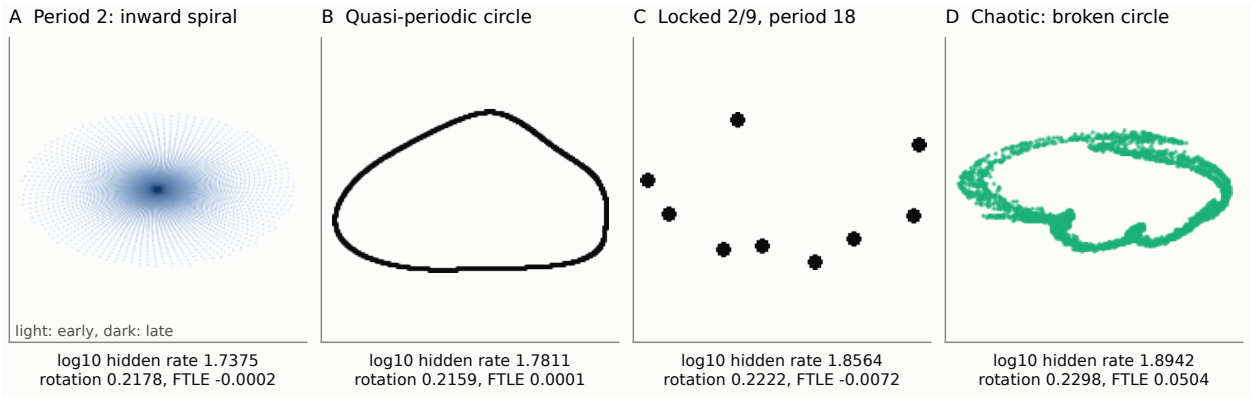


Figure 4: Four states along one slice, in order of increasing hidden-layer rate: an inward spiral to the 2-cycle, a closed invariant curve, nine locked points and a broken curve. Each panel shows every second parameter vector of one run projected on its own two principal axes. The rates are chosen by a fixed rule from the staircase receipt (the period-2 cell closest to the onset, the median quasi-periodic cell, the middle of the longest 2/9 plateau, the median chaotic cell between the 2/9 and 1/4 plateaus), not by eye. The states are not a monotone sequence: beyond the onset the rotation number first falls, and beyond panel D training locks again at 1/4 before chaos returns (Figure 5). Projections onto two axes can hide structure in the remaining dimensions. *Generated by* `paper/gd-route/figures/f6_route_portraits.py`; *receipt* `figures/f6_route_portraits.provenance.json`.

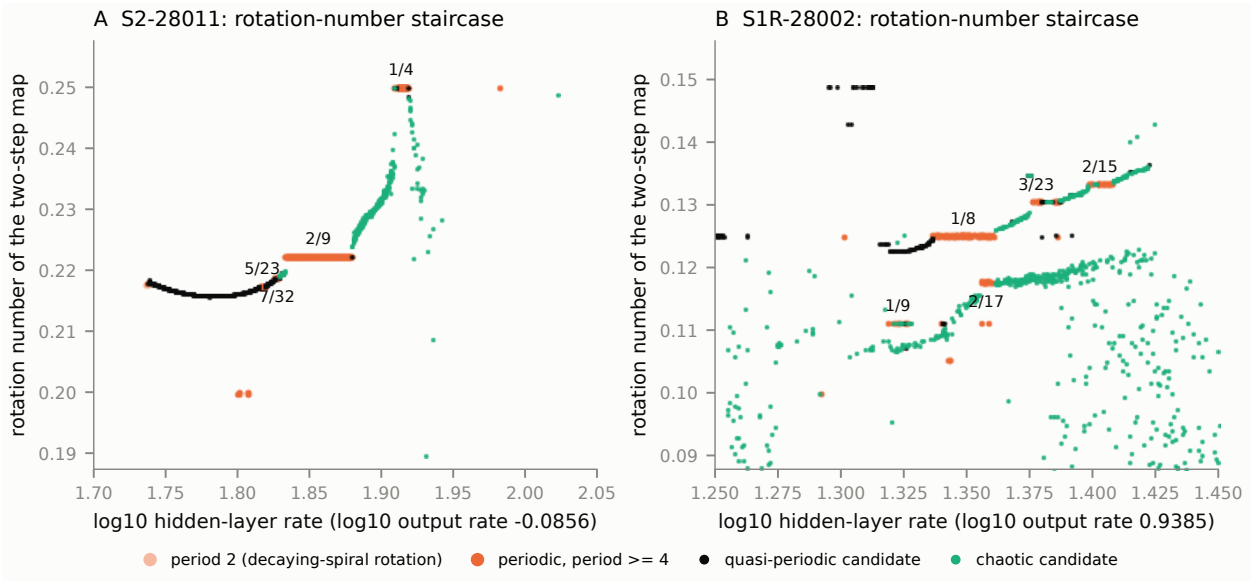


Figure 5: Along slices, quasi-periodic cells trace continuous rotation numbers and periodic cells sit on rational plateaus. Plateaus are detected automatically (locked cells with period $2q$ or a doubling of it, rotation within 6×10^{-4} of m/q) and labeled when they hold at least 8 cells. In S2 the rotation number first falls after the onset, then rises; the cells between the 2/9 and 1/4 plateaus are almost all chaotic candidates. In S1R two rotation branches coexist over the same rates, a sign of coexisting attractors. Chaotic cells outside the plotted range are not shown. *Generated by* `paper/gd-route/figures/f3_staircases.py`; *receipt* `figures/f3_staircases.provenance.json`.

then locks at 1/4 from 1.909 to 1.919 before chaos returns. The labeled plateaus are 5/23, 7/32, 2/9 and 1/4 in S2 and 1/9, 2/17, 1/8, 3/23 and 2/15 in S1R. Consecutive plateaus are Farey neighbors ($|m_1q_2 - m_2q_1| = 1$) in 3 of 3 pairs in S2, against 49% per pair for randomly drawn fractions in the same range, and in 4 of 4 pairs in S1R, where the null is 100% because every fraction with a small enough denominator in that range appears. The Farey pattern is weak evidence in S2 and uninformative in S1R.

5.5 A Birkhoff ordering test of the windows

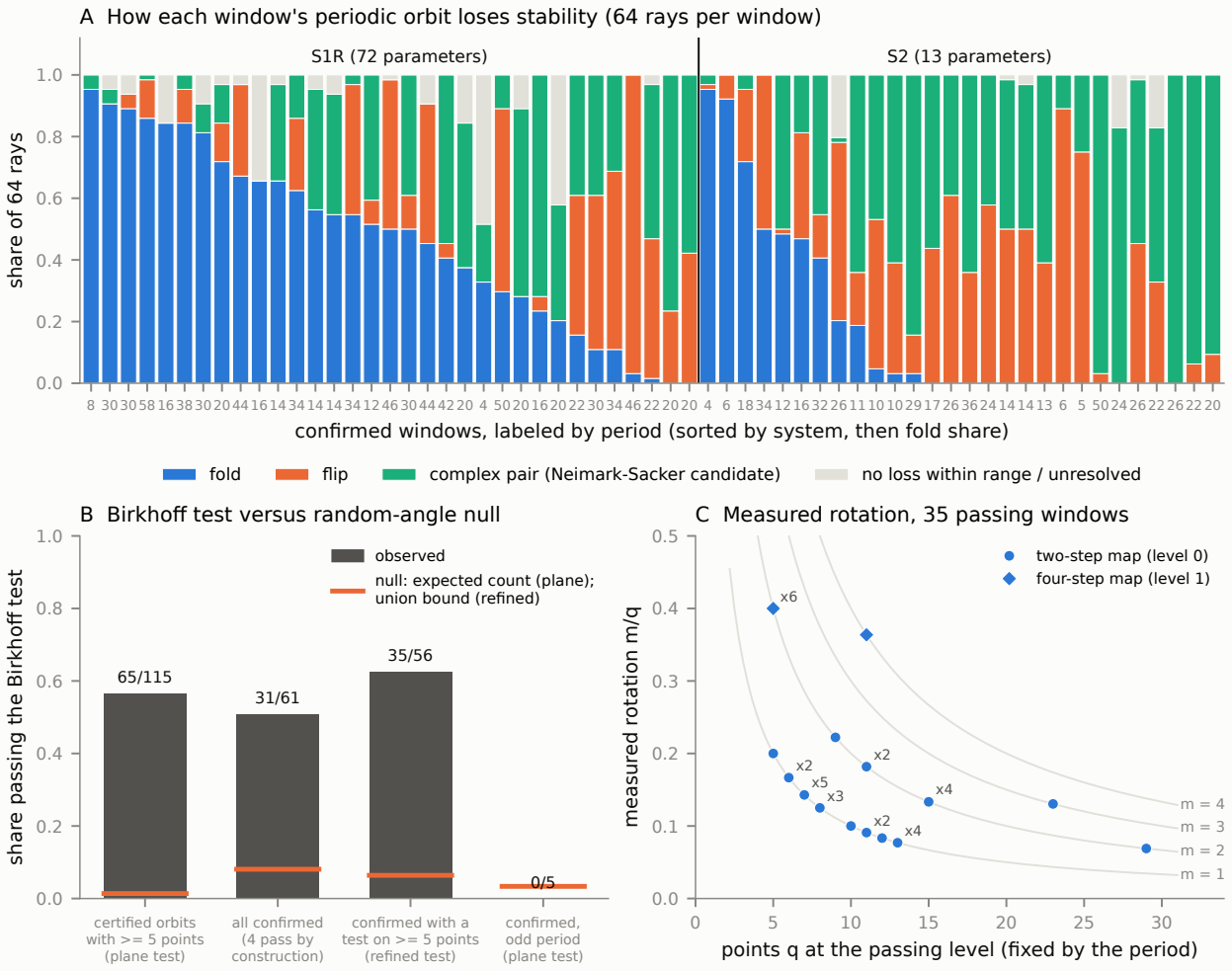


Figure 6: Across all 61 windows, stability boundaries mix fold, flip and complex-pair crossings, and most eligible windows pass a Birkhoff ordering test far above its null. A: boundary type along 64 rays per window. B: share of orbits passing the test (gray) against the null (orange): the expected share of chance passes for the plane test, and the union bound for the refined test; eligibility is printed under each bar. C: rotation m/q measured by the passing test at its first passing level; q is fixed by the period and level by construction, so only the shift m and the ordering are measured. *Generated by* `paper/gd-route/figures/f5_window_census.py`; *receipt* `figures/f5_window_census.provenance.json`.

Across the 61 windows, the 3,904 rays end at 1,316 folds, 1,012 flips and 1,395 complex-pair crossings (181 rays reached the end of their range or stayed unresolved; Figure 6A). Fold edges are what the edges of a locked window on a circle look like, while flips and secondary complex-pair

crossings are expected where a locked orbit loses stability off the circle.

The Birkhoff test results depend on which orbits are eligible, so we give each denominator. Of the 129 certified orbits with period at least 3, 115 have at least 5 two-step points; 65 of these pass the plane test, against 1.57 expected by chance. All 61 confirmed windows are eligible for the plane test and 31 pass, against 4.92 expected; this count includes 4 windows with at most 3 points, which pass by construction. For the refined test, 56 confirmed windows have at least one test on 5 or more points (the 5 excluded windows have periods 4, 4, 6, 6 and 8); 35 of them pass, against a union bound of 3.58 chance passes. 28 pass at the two-step level, with measured shifts $m = 1$ for 19, $m = 2$ for 8 and $m = 3$ for 1; 7 pass first at a period-doubled level, for example period-20 orbits ordered under four steps with rotation 2/5, twice 1/5. Of the 5 odd-period windows, 0 pass (null 0.17): they did not pass this diagnostic, and their organization is open.

5.6 Resonance ancestry of a rule-selected window: not established

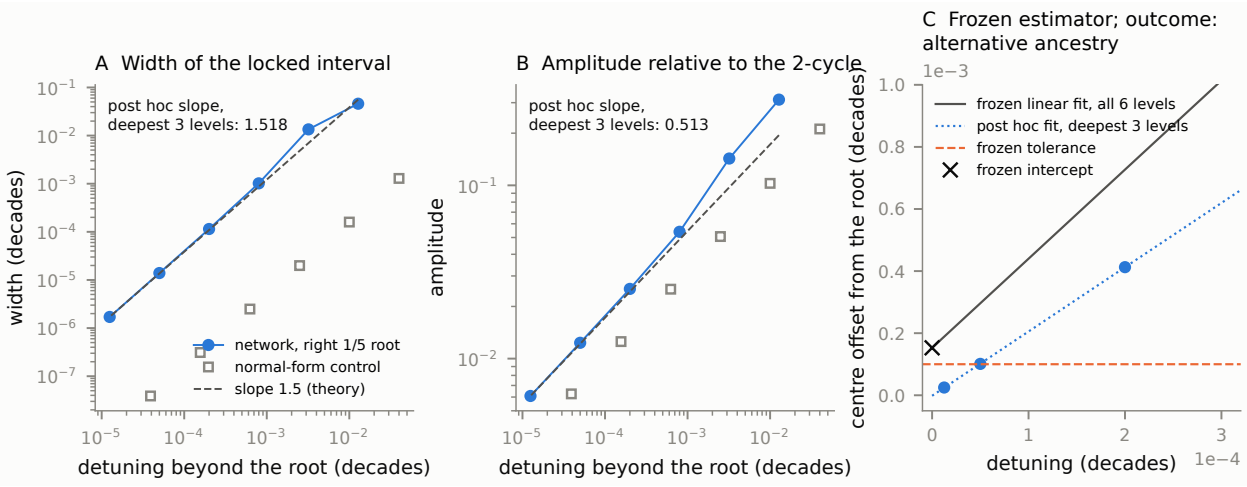


Figure 7: Near the predicted 1/5 resonance point, a fold-bounded locked region narrows with the scaling of a resonance tongue, but the frozen estimator places its tip outside the tolerance. A and B: width and amplitude of the locked period-10 interval on six cross-sections against the distance beyond the resonance point, with the resonant normal-form control and the theory slopes; the slopes printed are post hoc fits over the three deepest levels. C: the frozen estimator (linear fit of the interval's center over all six levels) and its intercept against the frozen tolerance, with a post-hoc fit over the deepest three levels; the axes show the deepest three levels. *Generated by* `paper/gd-route/figures/f8_ancestry.py`; *receipt* `figures/f8_ancestry.provenance.json`.

The test asks whether the rule-selected 1/5 window of S2-28011 descends from the 1/5 resonance of the crossing curve. The controls behaved as required: a resonant normal form's 1/5 tongue passed, the same branch scored against a displaced root was refused (alternative ancestry), and a logistic period-5 window born by a saddle-node was refused (alternative ancestry). At the right 1/5 resonance point 6 of 6 onset criteria passed (complex pair, other multipliers inside, transverse crossing, attracting circle with radius exponent 0.503, converging invariance residual, rotation equal to the crossing angle); at the left point 5 of 6 passed, the radius exponent 0.829 lying outside the frozen band.

From the right point, the locked interval has two fold edges on all 6 cross-sections, and its width and amplitude shrink toward the point (Figure 7). The frozen estimator nevertheless extrapolates the center to 1.5×10^{-4} decades from the predicted point, 53% above the tolerance of 10^{-4}, so the

frozen outcome is alternative ancestry. The connection from the tracked interval to the window ended in exhausted budget, and the left point gave no qualifying window. Under the frozen rules, the ancestry of this window is not established. Post hoc, and not replacing that outcome: the two shallowest levels lie outside the asymptotic regime, and over the three deepest levels the width scales with exponent 1.518 and the amplitude with 0.513, against $(q-2)/2 = 1.5$ and 0.5 for a Neimark–Sacker resonance tongue with $q = 5$.

5.7 Classical shrimp anatomy is the exception

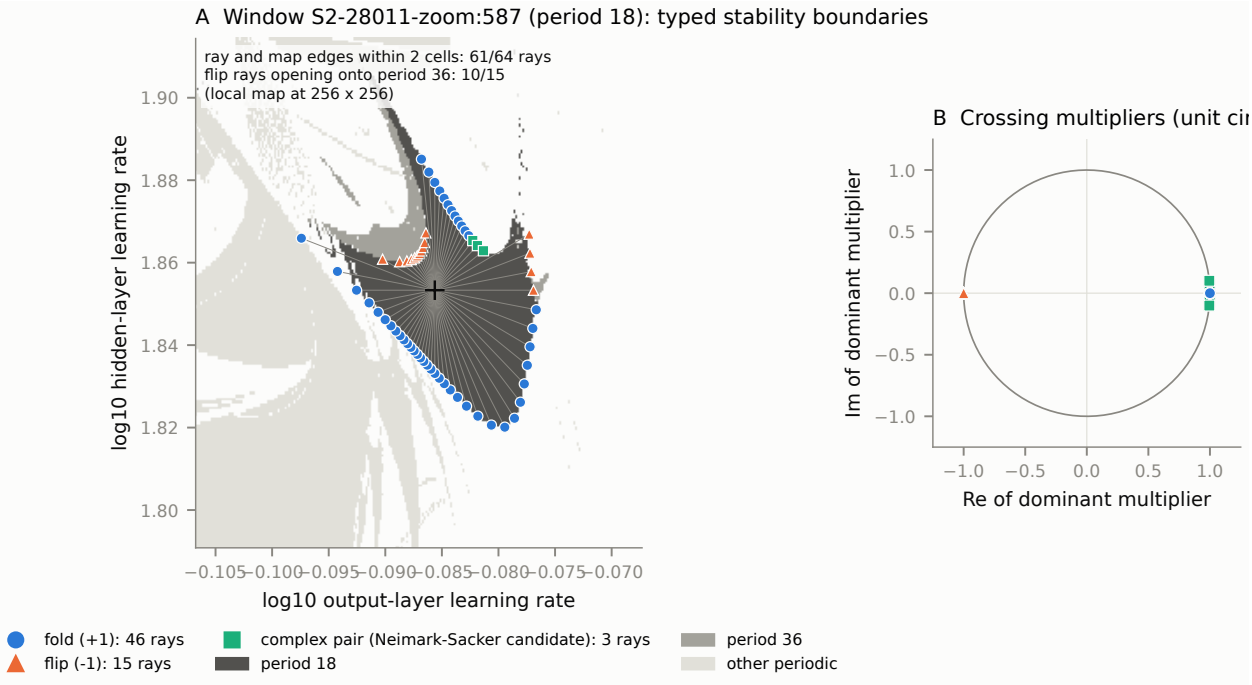


Figure 8: The period-18 window of S2-28011 has a fold-bounded body with flip edges opening onto period-36 bands. A: local map from the shared initial parameters with the window's orbit continued along 64 rays; markers show where and how it loses stability. B: crossing multipliers on the unit circle. This window was selected as the clearest example and is one of only 2 meeting the draft shrimp criteria; it is not representative (Figure 6). *Generated by* `paper/gd-route/figures/f4_window_anatomy.py`; *receipt* `figures/f4_window_anatomy.provenance.json`.

Draft criteria for classical shrimp anatomy require fold and flip boundaries each on at least 8 of 64 rays, at least half of the flip rays opening onto period $2p$ in a local map, and agreement between ray and map edges on at least three quarters of rays. 15 windows meet the boundary-count part and 2 meet all of it. The clearest, the period-18 window of S2-28011 (Figure 8), has 46 fold, 15 flip and 3 complex-pair rays; ray and map edges agree within two cells on 61 of 64 rays, and 10 of its flip rays open onto period 36. Complex-pair boundaries, absent from classical shrimps, are common: 41 of 61 windows have at least 8 such rays.

5.8 Beyond these two networks

Coexisting branches in S1R. The 72-parameter S1R network has several coexisting period-2 orbits, each with its own loss of stability. Of 36 output rates traced from one start, 16 end in a complex-pair crossing and 20 starts were not on a period-2 orbit; from a lower start, 25 of 36 end in

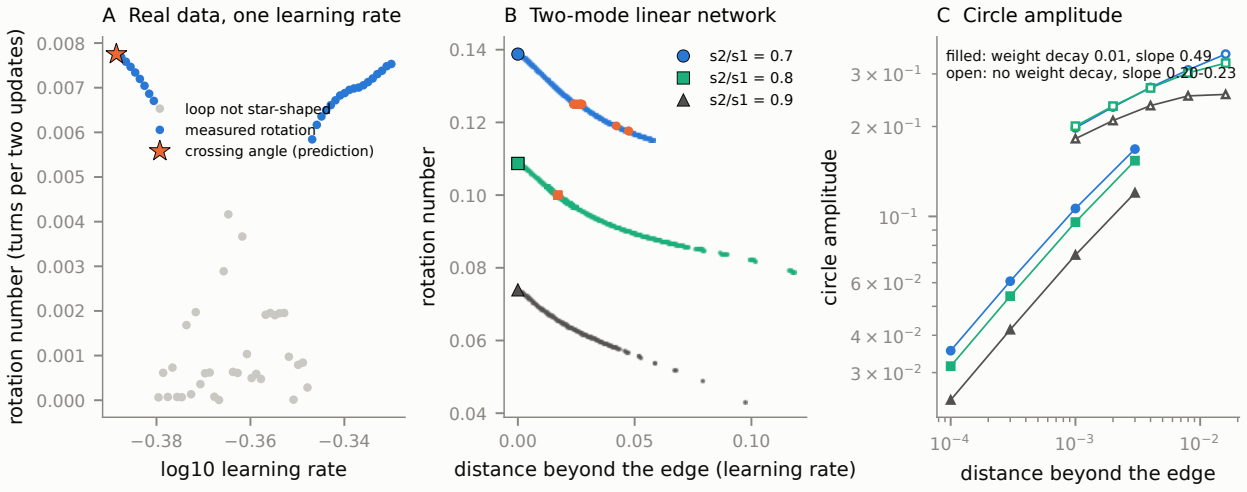


Figure 9: The same crossing, with the rotation predicted by its angle, in a real-data network with one learning rate and in Hofmann's two-mode linear network, where it is supercritical with locking beyond the edge. A: rotation number on the circle beyond the crossing of a 529-parameter tanh network on digit images (gray where the loop is not star-shaped about its centroid, so the angle estimate is invalid), and the crossing angle (star). B: rotation number against the distance beyond the rotational edge for three singular-value ratios, with weight decay; locked cells in orange. C: circle amplitude against the distance beyond the edge with and without weight decay; exponents are slopes between the two smallest offsets. *Generated by* `paper/gd-route/figures/f9_generality.py`; *receipt* `figures/f9_generality.provenance.json`.

a complex-pair crossing, 4 in folds, and 7 starts were not on a period-2 orbit. The two starts often reach different branches with different crossing angles.

Real data, one learning rate. A 64-8-1 tanh network (529 parameters) trained with one global learning rate and weight decay on 1,024 standardized UCI digit images (two-class task) has a certified period-2 orbit that loses stability at $\log_{10}\eta = -0.3885$ with dominant two-step multiplier 0.998814 + 0.048693i, a complex pair on the unit circle. Its angle predicts the rotation of the circle beyond: 0.0077528 turns per two updates, against 0.007751 measured just beyond the crossing on a smooth invariant circle (FTLE -0.0006; Figure 9A). This is one branch among those examined: of 4 certified period-2 branches continued to their loss of stability, 1 ends in a complex-pair crossing and 3 in folds, and on a second data subset Newton did not certify a period-2 orbit from 2 starts. No locked window was resolved; at rotations near 0.008 the relevant tongues have very large denominators.

Hofmann's rotational edge is a Neimark–Sacker bifurcation. In the two-mode depth-two linear network of Hofmann [13], the aligned 2-cycle loses stability at his rotational edge $a(s_1+s_2) = 2$, which we reproduce to 2.2×10^{-16} for $s_2/s_1 = 0.7, 0.8, 0.9$, with crossing angles 0.1380, 0.1082 and 0.0737 and a transverse crossing ($d|\mu|/da = 1.85$–1.95). Beyond the edge, training settles on an attracting invariant circle of the two-step map. Without weight decay the second mode's minima form a neutral continuum and the circle amplitude grows with exponent 0.20–0.23, a degenerate bifurcation; with weight decay 0.01 the exponent is 0.49, the square-root growth of a supercritical Neimark–Sacker bifurcation, and the rotation number at an offset of 10^{-4} is within 3.3×10^{-5} of the crossing angle (Figure 9B and C). Beyond the edge the rotation locks at 1/8 (period 16), 2/17 (period 34), 5/42 (period 84) for $s_2/s_1 = 0.7$, and at 1/10 (period 20) in a fine scan at 0.8. The same crossing occurs in the two-weight scalar factorization $\frac{1}{2}(uv - d)^2 + \frac{1}{2}\lambda(u^2 + v^2)$ with $\lambda = 0.2$ and unequal layer rates (crossing angle 0.4288; the circle amplitude grows by a factor 2.08 for a fourfold

offset, close to the square-root value 2), found as one of 6 of 64 cases of a coupled four-weight family that ended in a complex-pair crossing; boundary types for two held-out couplings were predicted correctly in 32 of 32 cases.

5.9 Consequences for training

Outcome (median per regime)	period 2 (282 cells)	quasi-periodic (125 cells)	locked (97 cells)	chaotic (223 cells)	ΔR^2 [95%]
function disagreement, 40k updates, start 10^{-8} apart ($\log_{10}$)	−15.87	−6.11	−14.69	−0.63	+0.000 [−0.179, +0.067]
function disagreement after the step-decay anneal ($\log_{10}$)	−15.81	−8.70	−9.41	−2.34	−0.005 [−0.243, +0.070]
spread over 64 anneal start times ($\log_{10}$)	−4.98	−3.73	−3.24	−2.36	+0.045 [+0.001, +0.107]
training loss, last 8,192 updates ($\log_{10}$)	−2.20	−2.10	−2.04	−1.93	−0.001 [−0.010, +0.001]
held-out loss, last 8,192 updates ($\log_{10}$)	−2.06	−1.94	−1.86	−1.73	−0.001 [−0.007, +0.001]
held-out loss after the anneal ($\log_{10}$)	−2.98	−2.98	−2.99	−3.05	+0.057 [−0.169, +0.232]
loss power in spectral lines	1.00	0.99	0.96	0.10	+0.001 [−0.037, +0.010]

Table 2: Consequences per regime (exploratory, S2-28011, one slice of 750 cells). Columns give medians per regime (cell counts in parentheses). ΔR^2 is the cross-validated gain from adding locking and rotation features (locked and quasi-periodic indicators, rotation number, distance to the nearest rational, $1/q$) to a base model of FTLE, sharpness, rate and oscillation amplitude, over the 729 cells beyond the edge of stability, with moving-block bootstrap intervals. Regime differences are large; the incremental value of rotation information is not.

The regime matters for training, but what it predicts is largely generic (Table 2). On a slice of 750 cells through the period-18 window, perturbations of the initial parameters die out on period-2 and most locked orbits, persist in proportion to their size on invariant circles, and saturate in chaos; locked windows reached through chaotic transients split into runs that reproduce exactly and runs that land on another phase of the same cycle. These differences follow the sign of the Lyapunov exponent, transient amplification, oscillation amplitude, sharpness and rate. Adding locking and rotation features to a model of those quantities gives an incremental R^2 whose interval lies above zero for 1 of 7 outcomes (the spread over anneal start times). In an analysis pooled over two slices with slice-specific slopes, changed after seeing an earlier fit and therefore adapted, two locking effects remain: the dose-response of disagreement on the perturbation size ($\Delta R^2 = +0.179$, [+0.070, +0.357]) and the spread over anneal start times (+0.047, [+0.011, +0.099]); both are carried by the locked indicator rather than the rotation number.

Reading the rotation from the loss. The dominant peak of the even-step training-loss spectrum equals the two-step rotation number within 10^{-3} in 125 of 125 quasi-periodic and 97 of 97 locked cells of the S2 slice, and in 31 of 31 and 19 of 19 on the S1R slice. Before the crossing, with small noise added to every update, an autoregressive fit of the even-step loss recovers the angle of the period-2 orbit's complex multiplier with median error 1.9×10^{-4} over 41 period-2 cells of S2 (largest 9.4×10^{-4}) and 3.6×10^{-4} over 34 cells of S1R; the estimator was chosen after a first pass (adapted). We treat this as local system identification, not early warning, until sampling, harmonic, abstention, autoregressive-baseline and specificity rules are fixed in advance.

Placeholder. Prespecified route-to-chaos path: a connected path, fixed in advance, from the period-2 region through the quasi-periodic band and locked windows into sustained chaos with converged diagnostics (not yet run).

Placeholder. Bounded transfer to real data: a small network on a CIFAR subset, developed on a first subset with everything frozen before a second subset is opened (not yet run).

6 Discussion

Everything in this section is interpretation. The confirmed results in Section 4 stand on their own, and the hypotheses here need their own prespecified tests.

The local part of the thesis now rests on several independent kinds of evidence: the crossing multipliers, the normal-form coefficient and its dynamical checks, the continuity of the rotation number with the crossing angle in two maps, and the same behaviour in a real-data network and in a solvable linear network. The picture that fits is the quasi-periodic organization familiar from circle maps and forced oscillators [3, 8, 15]: the edge-of-stability oscillation is a period-2 orbit; past a supercritical crossing, training moves on an invariant circle with a rotation number set by the crossing angle; rational rotations lock; and the circle breaks. The global part, that the periodic windows of the learning-rate plane are resonance tongues of this bifurcation, is supported by the Birkhoff test and by the post-hoc tongue scaling, but its frozen test did not pass. The frozen estimator mixed non-asymptotic and asymptotic levels; a confirmatory version should prespecify an asymptotic-regime estimator, and connecting a tongue to a distant window needs two-parameter fold-curve continuation.

The earliest spectral diagnostic of learning dynamics we know of is that of van der Maas et al. [32], who used broadband spectra to recognize chaotic learning; here a spectral peak carries the rotation number, and an autoregressive fit carries the complex multiplier before the crossing.

Open questions. The odd-period windows did not pass the Birkhoff diagnostic; whether they are organized as in periodicity hubs [17] is open. Fixed-start maps mix local criticality with basin accessibility from one shared initialization, as the divergent arc shows, so statements about trainability maps need both. In the real-data network only one of the examined period-2 branches crossed by complex multipliers, and how common such branches are in larger networks is unknown.

The tests that would settle the local claims. The criticality prediction is ready to freeze for fresh losses: the sign of d at each crossing predicts a circle of normal-form amplitude beyond a supercritical edge, and a jump with square-root escape thresholds and hysteresis at a subcritical one. An existence-and-replication test would apply this, together with the onset and rotation criteria, to two prespecified independent losses across two architectures, one of them unused so far, with no substitution. The near-root ancestry test is close to ready once its estimator is restricted to the asymptotic levels.

7 Limitations

The map study uses networks of 13 and 72 parameters on synthetic data; the real-data result is one branch of one 529-parameter network on one subset and task; the linear network is a solvable model. Training is deterministic, full batch and in float64. Screening labels are finite-time; chaotic labels rest on single-tangent exponents and reruns. Certification is a floating-point verification. The first

Lyapunov coefficient is a local, third-order quantity; it predicts the birth of a small circle, not how far it survives. Rotation numbers and the Birkhoff test use two-dimensional projections of curves in many dimensions. Most mechanism analyses were chosen after seeing related outcomes. No result addresses generalization, training speed or modern training practice.

8 Related work

Edge of stability. Cohen et al. [6] describe the edge of stability; Damian et al. [9] explain self-stabilization of sharpness at the edge, whereas we follow what happens after the period-2 oscillation itself loses stability; Agarwala et al. [1] show progressive sharpening to the edge in second-order regression models, the approach to the regime we study beyond it; Lewkowycz et al. [21] describe a catapult phase at large rates, in which the loss first grows and then settles in flatter regions, while our maps resolve the long-run dynamics of rates where training does not settle. Hofmann [13] derives the rotational edge whose far side we study. Further analyses include Kalra et al. [18], Chen and Bruna [4], Song and Yun [29], Chen et al. [5] and Cohen et al. [7].

Chaos and fractals in training. Chaos in deterministic training is discussed by Kong and Tao [19] and Jiménez-González et al. [16]; fractal trainability boundaries by Sohl-Dickstein [28], Torkamandi [31], Liu [23] and Liang and Montúfar [22]; riddled basins and reproducibility by Ly and Gong [24]. Learning dynamics in games show chaos [27] and phase locking [25], for adaptive rules rather than descent on one loss.

Dynamical systems. Shrimp-shaped windows are described by Gallas [10, 11]; circle maps, Arnold tongues, devil's staircases and the breakdown of invariant circles by Arnold [2], Aronson et al. [3], Curry and Yorke [8], Jensen et al. [14, 15], Ostlund et al. [26]; numerical bifurcation analysis of maps, including the normal-form coefficients used here, by Kuznetsov and Meijer [20]. A literature check using generic queries found no report of quasi-periodic training, Arnold tongues or rotation-number staircases for optimizers acting on one loss.

9 Reproducibility

Code, plans and compact receipts live in the project repository; bulky raw maps live in a separate data store referenced by SHA-256. Every figure is generated by a script in `paper/gd-route/figures/` that reads saved evidence and writes a receipt with input hashes, the generator's hash, output hashes, alt text and the values it displays. Every number in the text, and both tables, are generated by `paper/gd-route/scripts/build_numbers.py`; `data/numbers.json` records each number's source file, hash and computation, and `provenance.json` indexes all receipts. `paper/gd-route/scripts/verify.py` regenerates every figure and number and fails on any difference. The verification status of every reference is kept in `data/references-ledger.json`. The map study corresponds to the repository's EXP-028 plan and audit; its plan, receipts and audit are linked from `docs/EXP-028-results.md`. The exploratory streams are summarized in `docs/explore-criticality-2026-10-07.md`, `docs/explore-ancestry-2026-10-07.md`, `docs/explore-realdata-2026-10-07.md`, `docs/explore-minimal-model-2026-10-07.md` and `docs/explore-consequence-2026-10-07.md`.

Placeholder. Author contributions, acknowledgments, license choice and public code and data links are human decisions to be made before release.

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